

Global Healthcare: Pharmaceuticals: Lung Cancer KOLs Generally Positive on ASCO Developments

On the back of the [American Society of Clinical Oncology \(ASCO\) conference](#), we hosted two lung cancer KOLs to discuss key updates, notably MRK/Kelun's sac-TMT [optiTROP-05 trial in NSCLC](#), SMMT/Akeso's single-region (China) [Ph3 HARMONi-6 overall survival results](#) for ivonescimab+chemotherapy in frontline squamous NSCLC, and readthrough on programs from AZN, BNTX/BMY, and PFE.

Key takeaways from the discussion include: 1) sac-TMT's strong PFS was well-received with the KOLs optimistic around the safety profile and ability to manage toxicity, noting the final global overall survival (OS) data will ultimately be important in guiding treatment decisions; 2) ivo's OS benefit over Tevimbra (anti-PD-1, similar to MRK's Keytruda) is meaningful, albeit the KOLs acknowledged debates regarding efficacy in older patients and noted confirmation in the global Ph3 HARMONi-3 study is needed, where they see a high PoS for the squamous cohort final PFS analysis in 2H; and 3) For AZN's Datroway, while the developmental lead was noted, the KOLs see potential for superior efficacy and safety from sac-TMT based on China data, contingent on global trial validation. The KOLs also highlighted novel approaches they are watching in lung cancer including tri-specifics (PD1xCTLA4xVEGF) and RAS-inhibitors.

Asad Haider, CFA
+1(212)902-0691 | asad.haider@gs.com
Goldman Sachs & Co. LLC

Salveen Richter, CFA
+1(212)934-4204 | salveen.richter@gs.com
Goldman Sachs & Co. LLC

Rajan Sharma
+44(20)7051-7995 | rajan.sharma@gs.com
Goldman Sachs International

Nick Jennings
+1(415)249-7412 | nick.jennings@gs.com
Goldman Sachs & Co. LLC

Mark Aleynick, Ph.D.
+1(212)357-6820 | mark.aleynick@gs.com
Goldman Sachs & Co. LLC

Max Da, Ph.D.
+44(20)7051-8835 | max.da@gs.com
Goldman Sachs International

Jeff Su
+1(212)357-9930 | jeff.su@gs.com
Goldman Sachs & Co. LLC

Key Takeaways

In parallel with the American Society of Clinical Oncology (ASCO) meeting, we spoke with two KOLs: 1) a Professor of Oncology at Johns Hopkins University School of Medicine and Trinity College Dublin, and 2) an Assistant Professor, and Medical Director of Oncology at Stanford University, to discuss key non-small cell lung cancer (NSCLC) data from the conference. The Johns Hopkins KOL is a thoracic medical oncologist who sees ~200 patients annually in his practice, almost all NSCLC, with a ~70% non-squamous, 25%-30% squamous distribution, and a focus on early-phase immunotherapy trials. The Stanford KOL is a medical oncologist who sees a substantial NSCLC population, with prior experience in gastrointestinal cancer and melanoma via prior roles at Memorial Sloan Kettering Cancer Center and the Dana-Farber Cancer Institute.

We highlight the following key takeaways from the discussion:

- **Current 1L mNSCLC patient management.** Both doctors first check for the

Goldman Sachs does and seeks to do business with companies covered in its research reports. As a result, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of this report. Investors should consider this report as only a single factor in making their investment decision. For Reg AC certification and other important disclosures, see the Disclosure Appendix, or go to www.gs.com/research/hedge.html. Analysts employed by non-US affiliates are not registered/qualified as research analysts with FINRA in the U.S.

presence of actionable genetic alterations, and if there are none, treatment is determined by PD-L1 expression and histology (squamous/non-squamous). For patients that are PD-L1 TPS $\geq$ 50%, one KOL favors Keytruda monotherapy while the other favors Keytruda + chemotherapy, because he feels that provides the best opportunity for a response. For patients below TPS 50%, both use Keytruda + chemo, with the chemo being histology specific.

- **Overall impressions of MRK partner Kelun's sac-TMT OptiTROP-Lung05 data.** From an efficacy and safety perspective, one KOL noted the sac-TMT arm performance was impressive (one of the few drugs that he has seen show a PFS HR that significant, acknowledging that Keytruda monotherapy is not the relevant control arm globally) and that the tox is a little more than chemo so is watching AEs that can be an issue for some patients, such as stomatitis and ILD, but overall is optimistic that the global trials will be positive. The other KOL noted the Keytruda arm seemed to underperform, and there was a trend to OS, and described the sac-TMT arm discontinuation rate of ~5% as amazing.
- **sac-TMT toxicity profile.** Focusing on the safety data known thus far, one KOL noted the profile seems similar to AZN's Dato-DXD, and notes that in his experience with ADCs, patients' individual reactions and/or ability to tolerate can be highly variable. The other KOL noted that when he discusses ADCs with patients, he anticipates chemo-like side effects, and believes those to be manageable, and noted that he was encouraged that they did not see unique side effects (pneumonitis, ocular, etc).
- **sac-TMT efficacy profile.** From an efficacy perspective, the KOLs agreed that all groups showed benefit, with PD-L1 high where there is clearest relevance (given Keytruda monotherapy control arm), as there are some patients for whom Keytruda mono is the better choice.
- **Bars for clinical decision making.** One KOL provided a range for how he interprets HRs in baseball terms, with better than 0.85 being a single, ~0.7 to ~0.85 being a double, ~0.5 to ~0.7 being a triple and below 0.5 being a home run, and also considering the shape of the curve, aiming to steer patients towards options with higher survival benefit — for the next sac-TMT data set, he would be looking for HR of 0.7 to 0.8 with response rates of 65% to 70% to displace the standard of care, noting the Keytruda patent expiry in 2028 could increase pressure (by insurers) to use the drug. The other KOL agrees broadly, but sees ADCs more for non-squamous disease, and PD1xVEGF for squamous — for OS, he notes that in order to be stat sig, HR will usually be less than 0.8, and believes the bar for clinically meaningful HR is 0.75.
- **Overall impressions of SMMT partner Akeso's HARMONi-6 OS data.** The KOLs noted the overall survival (OS) benefit seen with SMMT and partner Akeso's PD1xVEGF bispecific ivonescimab (ivo) over Tevimbra (ONC's anti-PD-1, similar to MRK's Keytruda) + chemotherapy in the Ph3 HARMONi-6 study is meaningful and suggests ivo is an effective therapy, with relatively little drop-off between progression-free survival (PFS; HR=0.60) and OS (HR=0.66). **However, acknowledging debates from the plenary discussion section regarding the**

perceived lack of benefit in the age>65 subgroup ([see management commentary here](#)), they cautioned that the China-only study needs to be repeated in a global population (which typically includes older patients).

Additionally, regional pathophysiological differences in NSCLC per smoking and other factors is possible, further supporting the need for global confirmation. On safety, the KOLs highlighted that HARMONi-6 included patients who have higher bleeding risk per blood vessel encasement, who would traditionally not be candidates for VEGF therapies, noting relatively little VEGF-mediated toxicity (hypertension, hemorrhage) for ivo as differentiating from bevacizumab, and that immune-mediated toxicities were similar to those seen with Keytruda.

- **HARMONi-6 translation to the global HARMONi-3 study.** On translation of the HARMONi-6 results to SMMT's global Ph3 HARMONi-3 frontline NSCLC study, the KOLs believe the final PFS results from the squamous HARMONi-3 cohort in 2H are likely to be positive, as the totality of positive ivo data across HARMONi-6 and other indications (e.g. [global Ph2 colorectal cancer data shared at ASCO](#)) suggests the efficacy signal is real. When discussing readthrough of the squamous-only HARMONi-6 data to non-squamous NSCLC (recall, the global Ph3 HARMONi-3 study contains a squamous and a non-squamous NSCLC patient cohort), the KOLs noted that squamous NSCLC is generally more homogeneous, with a greater prevalence in smokers, thus per the available data greater benefit is expected in squamous patients, noting modest readthrough between the histologies. Overall, the KOLs believe ivo could potentially be better than Keytruda, and await data from the global Ph3 HARMONi-3 study for clarity on the magnitude of benefit.
- **When comparing NSCLC data across bispecifics, including ASCO presentations from BNTX/BMY's pumitamidg and PFE's PF'4404, the KOLs noted minimal drop-off in efficacy when going from China-only to global data,** and that the data overall support the PD1/L1xVEGF class. The KOLs additionally highlighted promising novel approaches including the tri-specific PD1xCTLA4xVEGF molecule CS2009 from the Chinese biotech CStone Pharmaceuticals.
- **The KOLs highlighted AZN's Datroway's developmental lead while acknowledging sac-TMT's potential for superior efficacy and safety based on China data, contingent on global trial validation.** The KOLs highlighted Datroway's developmental lead, noting the FDA approval of the monotherapy and established clinical experience, which simplifies physicians' integration of the combination of Datroway and IO therapies into their treatment algorithm. While acknowledging sac-TMT's potential for superior efficacy, based on current China data, and its low discontinuation rates in those trials, the KOLs underscored the requirement for these results to be validated in global studies. With respect to the AVANZAR trial, the biomarker strategy was noted as sound, with Datroway showing potential for greater efficacy over pemetrexed. However, the selection of Imfinzi instead of Keytruda for the control arm introduced risks.
- **Novel approaches the KOLs are watching in lung cancer.** In terms of emerging treatment paradigms, the KOLs highlighted potential for multiple "novel-novel" combinations including ADCs + IO, bispecifics, trispecifics (PD1xCTLA4xVEGF) and called out RAS-inhibitors as an area of interest (including combining these with

IOs/ADCs).

Valuation and Risks

MRK (Buy). Valuation: We arrive at our 12-month target of \$137 using a 14.0x multiple on our (Q5-Q8) adjusted EPS. Downside risks: Faster-than-expected erosion of the Keytruda LOE / lower than expected conversion to SubQ Keytruda. Further slowdown in established vaccine products, slower ramp of newly launched products, and regulatory uncertainty. Disappointments from pipeline assets. Tougher-than-expected regulatory/market environment related to vaccines, FDA, and tariffs, and M&A.

SMMT (Buy). Valuation: Our 12-month PT of \$41 is based on a blended 85%/15% risk-adjusted DCF value of \$35 (WACC of 18% and TGR of 2%) and a theoretical M&A value of \$73 (11x 2031E sales). Downside risks: Clinical, technology and manufacturing risk - data generated by partner Akeso in China may not translate to SMMT's global clinical studies; U.S.-China trade relations may adversely impact supply chain operations; unexpected safety issues or clinical delays may push launch timelines. Regulatory risk - failure to secure regulatory approval for indications or territories could impact commercial success and would pose significant downside risk to our sales estimates. Competitive risk - competitor PD-1/L1xVEGF assets could demonstrate superior data, or advance through clinical development ahead of expectations. Financing risk/dilution.

AstraZeneca (Buy): Our 12-month price target is derived from a 50:50 blend of DCF and P/E. Our bottom-up DCF analysis suggests a valuation of 15,912p per share (WACC 8.0%; TGR 2.5%) and on a multiples basis we believe AstraZeneca should trade on 20x 2027E EPS, giving a valuation of 17,318p per share. As a result, our price targets are 16,525p/\$219 per share. Key risks to our view and price target: (i) Clinical trial failure; (ii) Commercial execution; (iii) Pricing; (iv) Competitive developments; and (v) Patent exposure.

Disclosure Appendix

Reg AC

We, Asad Haider, CFA, Salveen Richter, CFA, Rajan Sharma, Nick Jennings, Mark Aleynick, Ph.D., Max Da, Ph.D. and Jeff Su, hereby certify that all of the views expressed in this report accurately reflect our personal views about the subject company or companies and its or their securities. We also certify that no part of our compensation was, is or will be, directly or indirectly, related to the specific recommendations or views expressed in this report.

Unless otherwise stated, the individuals listed on the cover page of this report are analysts in Goldman Sachs' Global Investment Research division.

Contributing Authors: Asad Haider, CFA Goldman Sachs & Co. LLC, Salveen Richter, CFA Goldman Sachs & Co. LLC, Rajan Sharma Goldman Sachs International, Nick Jennings Goldman Sachs & Co. LLC, Mark Aleynick, Ph.D. Goldman Sachs & Co. LLC, Max Da, Ph.D. Goldman Sachs International, Jeff Su Goldman Sachs & Co. LLC.

Unless otherwise stated, the individuals listed in the Contributing Authors disclosure of this report are analysts in Goldman Sachs' Global Investment Research division.

GS Factor Profile

The Goldman Sachs Factor Profile provides investment context for a stock by comparing key attributes to the market (i.e. our universe of rated stocks) and its sector peers. The four key attributes depicted are: Growth, Financial Returns, Multiple (e.g. valuation) and Integrated (a composite of Growth, Financial Returns and Multiple). Growth, Financial Returns and Multiple are calculated by using normalized ranks for specific metrics for each stock. The normalized ranks for the metrics are then averaged and converted into percentiles for the relevant attribute. The precise calculation of each metric may vary depending on the fiscal year, industry and region, but the standard approach is as follows:

Growth is based on a stock's forward-looking sales growth, EBITDA growth and EPS growth (for financial stocks, only EPS and sales growth), with a higher percentile indicating a higher growth company. **Financial Returns** is based on a stock's forward-looking ROE, ROCE and CROCI (for financial stocks, only ROE), with a higher percentile indicating a company with higher financial returns. **Multiple** is based on a stock's forward-looking P/E, P/B, price/dividend (P/D), EV/EBITDA, EV/FCF and EV/Debt Adjusted Cash Flow (DAFCF) (for financial stocks, only P/E, P/B and P/D), with a higher percentile indicating a stock trading at a higher multiple. The **Integrated** percentile is calculated as the average of the Growth percentile, Financial Returns percentile and (100% - Multiple percentile).

Financial Returns and Multiple use the Goldman Sachs analyst forecasts at the fiscal year-end at least three quarters in the future. Growth uses inputs for the fiscal year at least seven quarters in the future compared with the year at least three quarters in the future (on a per-share basis for all metrics).

For a more detailed description of how we calculate the GS Factor Profile, please contact your GS representative.

M&A Rank

Across our global coverage, we examine stocks using an M&A framework, considering both qualitative factors and quantitative factors (which may vary across sectors and regions) to incorporate the potential that certain companies could be acquired. We then assign a M&A rank as a means of scoring companies under our rated coverage from 1 to 3, with 1 representing high (30%-50%) probability of the company becoming an acquisition target, 2 representing medium (15%-30%) probability and 3 representing low (0%-15%) probability. For companies ranked 1 or 2, in line with our standard departmental guidelines we incorporate an M&A component into our target price. M&A rank of 3 is considered immaterial and therefore does not factor into our price target, and may or may not be discussed in research.

Quantum

Quantum is Goldman Sachs' proprietary database providing access to detailed financial statement histories, forecasts and ratios. It can be used for in-depth analysis of a single company, or to make comparisons between companies in different sectors and markets.

Disclosures

Pricing information

AstraZeneca (\$179.71), AstraZeneca (13,434p), Merck & Co. (\$115.17) and Summit Therapeutics Inc. (\$15.71)

Distribution of ratings/investment banking relationships

Goldman Sachs Investment Research global Equity coverage universe

	Rating Distribution			Investment Banking Relationships		
	Buy	Hold	Sell	Buy	Hold	Sell
Global	50%	34%	16%	65%	60%	45%

As of April 1, 2026, Goldman Sachs Global Investment Research had investment ratings on 3,074 equity securities. Goldman Sachs assigns stocks as Buys and Sells on various regional Investment Lists; stocks not so assigned are deemed Neutral. Such assignments equate to Buy, Hold and Sell for the purposes of the above disclosure required by the FINRA Rules. See 'Ratings, Coverage universe and related definitions' below. The Investment Banking Relationships chart reflects the percentage of subject companies within each rating category for whom Goldman Sachs has provided investment banking services within the previous twelve months.

Regulatory disclosures

Disclosures required by United States laws and regulations

See company-specific regulatory disclosures above for any of the following disclosures required as to companies referred to in this report: manager or co-manager in a pending transaction; 1% or other ownership; compensation for certain services; types of client relationships; managed/co-managed public offerings in prior periods; directorships; for equity securities, market making and/or specialist role. Goldman Sachs trades or may trade as a principal in debt securities (or in related derivatives) of issuers discussed in this report.

The following are additional required disclosures: **Ownership and material conflicts of interest:** Goldman Sachs policy prohibits its analysts, professionals reporting to analysts and members of their households from owning securities of any company in the analyst's area of coverage. **Analyst compensation:** Analysts are paid in part based on the profitability of Goldman Sachs, which includes investment banking revenues. **Analyst as officer**

or director: Goldman Sachs policy generally prohibits its analysts, persons reporting to analysts or members of their households from serving as an officer, director or advisor of any company in the analyst's area of coverage. **Non-U.S. Analysts:** Non-U.S. analysts may not be associated persons of Goldman Sachs & Co. LLC and therefore may not be subject to FINRA Rule 2241 or FINRA Rule 2242 restrictions on communications with a subject company, public appearances and trading in securities covered by the analysts.

Distribution of ratings: See the distribution of ratings disclosure above. **Price chart:** See the price chart, with changes of ratings and price targets in prior periods, above, or, if electronic format or if with respect to multiple companies which are the subject of this report, on the Goldman Sachs website at <https://www.gs.com/research/hedge.html>.

Additional disclosures required under the laws and regulations of jurisdictions other than the United States

The following disclosures are those required by the jurisdiction indicated, except to the extent already made above pursuant to United States laws and regulations. **Australia:** Goldman Sachs Australia Pty Ltd and its affiliates are not authorised deposit-taking institutions (as that term is defined in the Banking Act 1959 (Cth)) in Australia and do not provide banking services, nor carry on a banking business, in Australia. This research, and any access to it, is intended only for "wholesale clients" within the meaning of the Australian Corporations Act, unless otherwise agreed by Goldman Sachs. In producing research reports, members of Global Investment Research of Goldman Sachs Australia may attend site visits and other meetings hosted by the companies and other entities which are the subject of its research reports. In some instances the costs of such site visits or meetings may be met in part or in whole by the issuers concerned if Goldman Sachs Australia considers it is appropriate and reasonable in the specific circumstances relating to the site visit or meeting. To the extent that the contents of this document contains any financial product advice, it is general advice only and has been prepared by Goldman Sachs without taking into account a client's objectives, financial situation or needs. A client should, before acting on any such advice, consider the appropriateness of the advice having regard to the client's own objectives, financial situation and needs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests and a copy of Goldman Sachs' Australian Sell-Side Research Independence Policy Statement are available at: <https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html>. **Brazil:** Disclosure information in relation to CVM Resolution n. 20 is available at <https://www.gs.com/worldwide/brazil/area/gir/index.html>. Where applicable, the Brazil-registered analyst primarily responsible for the content of this research report, as defined in Article 20 of CVM Resolution n. 20, is the first author named at the beginning of this report, unless indicated otherwise at the end of the text. **Canada:** This information is being provided to you for information purposes only and is not, and under no circumstances should be construed as, an advertisement, offering or solicitation by Goldman Sachs & Co. LLC for purchasers of securities in Canada to trade in any Canadian security. Goldman Sachs & Co. LLC is not registered as a dealer in any jurisdiction in Canada under applicable Canadian securities laws and generally is not permitted to trade in Canadian securities and may be prohibited from selling certain securities and products in certain jurisdictions in Canada. If you wish to trade in any Canadian securities or other products in Canada please contact Goldman Sachs Canada Inc., an affiliate of The Goldman Sachs Group Inc., or another registered Canadian dealer. **Hong Kong:** Further information on the securities of covered companies referred to in this research may be obtained on request from Goldman Sachs (Asia) L.L.C. **India:** Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (India) Securities Private Limited, Research Analyst - SEBI Registration Number INH000001493, 10th Floor, Ascent-Worli, Sudam Kalu Ahire Marg, Worli, Mumbai-400 025, India, Corporate Identity Number U74140MH2006FTC160634, Phone +91 22 6616 9000, Fax +91 22 6616 9001. Goldman Sachs may beneficially own 1% or more of the securities (as such term is defined in clause 2 (h) of the Indian Securities Contracts (Regulation) Act, 1956) of the subject company or companies referred to in this research report. Investment in securities market are subject to market risks. Read all the related documents carefully before investing. Registration granted by SEBI and certification from NISM in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Goldman Sachs (India) Securities Private Limited compliance officer and investor grievance contact details can be found at: <https://www.goldmansachs.com/worldwide/india/documents/Grievance-Redressal-and-Escalation-Matrix.pdf>, and a copy of the annual audit compliance report can be found at this link: <https://publishing.gs.com/content/site/india-annual-compliance-report.html>. **Japan:** See below. **Korea:** This research, and any access to it, is intended only for "professional investors" within the meaning of the Financial Services and Capital Markets Act, unless otherwise agreed by Goldman Sachs. Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (Asia) L.L.C., Seoul Branch. **New Zealand:** Goldman Sachs New Zealand Limited and its affiliates are neither "registered banks" nor "deposit takers" (as defined in the Reserve Bank of New Zealand Act 1989) in New Zealand. This research, and any access to it, is intended for "wholesale clients" (as defined in the Financial Advisers Act 2008) unless otherwise agreed by Goldman Sachs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests is available at: <https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html>. **Russia:** Research reports distributed in the Russian Federation are not advertising as defined in the Russian legislation, but are information and analysis not having product promotion as their main purpose and do not provide appraisal within the meaning of the Russian legislation on appraisal activity. Research reports do not constitute a personalized investment recommendation as defined in Russian laws and regulations, are not addressed to a specific client, and are prepared without analyzing the financial circumstances, investment profiles or risk profiles of clients. Goldman Sachs assumes no responsibility for any investment decisions that may be taken by a client or any other person based on this research report. **Singapore:** Goldman Sachs (Singapore) Pte. (Company Number: 198602165W), which is regulated by the Monetary Authority of Singapore, accepts legal responsibility for this research, and should be contacted with respect to any matters arising from, or in connection with, this research. **Taiwan:** This material is for reference only and must not be reprinted without permission. Investors should carefully consider their own investment risk. Investment results are the responsibility of the individual investor. **United Kingdom:** Persons who would be categorized as retail clients in the United Kingdom, as such term is defined in the rules of the Financial Conduct Authority, should read this research in conjunction with prior Goldman Sachs research on the covered companies referred to herein and should refer to the risk warnings that have been sent to them by Goldman Sachs International. A copy of these risks warnings, and a glossary of certain financial terms used in this report, are available from Goldman Sachs International on request.

European Union and United Kingdom: Disclosure information in relation to Article 6 (2) of the European Commission Delegated Regulation (EU) (2016/958) supplementing Regulation (EU) No 596/2014 of the European Parliament and of the Council (including as that Delegated Regulation is implemented into United Kingdom domestic law and regulation following the United Kingdom's departure from the European Union and the European Economic Area) with regard to regulatory technical standards for the technical arrangements for objective presentation of investment recommendations or other information recommending or suggesting an investment strategy and for disclosure of particular interests or indications of conflicts of interest is available at <https://www.gs.com/disclosures/europeanpolicy.html> which states the European Policy for Managing Conflicts of Interest in Connection with Investment Research.

Japan: Goldman Sachs Japan Co., Ltd. is a Financial Instrument Dealer registered with the Kanto Financial Bureau under registration number Kinsho 69, and a member of Japan Securities Dealers Association, Financial Futures Association of Japan Type II Financial Instruments Firms Association, and Investment Management Association of Japan. Sales and purchase of equities are subject to commission pre-determined with clients plus consumption tax. See company-specific disclosures as to any applicable disclosures required by Japanese stock exchanges, the Japanese Securities Dealers Association or the Japanese Securities Finance Company.

Ratings, coverage universe and related definitions

Buy (B), Neutral (N), Sell (S) Analysts recommend stocks as Buys or Sells for inclusion on various regional Investment Lists. Being assigned a Buy or Sell on an Investment List is determined by a stock's total return potential relative to its coverage universe. Any stock not assigned as a Buy or a Sell on an Investment List with an active rating (i.e., a stock that is not Rating Suspended, Not Rated, Early-Stage Biotech, Coverage Suspended or Not Covered), is deemed Neutral. Each region manages Regional Conviction Lists, which are selected from Buy rated stocks on the respective region's Investment Lists and represent investment recommendations focused on the size of the total return potential and/or the likelihood of the realization of the return across their respective areas of coverage. The addition or removal of stocks from such Conviction Lists are managed by the Investment Review Committee or

other designated committee in each respective region and do not represent a change in the analysts' investment rating for such stocks.

Total return potential represents the upside or downside differential between the current share price and the price target, including all paid or anticipated dividends, expected during the time horizon associated with the price target. Price targets are required for all covered stocks. The total return potential, price target and associated time horizon are stated in each report adding or reiterating an Investment List membership.

Coverage Universe: A list of all stocks in each coverage universe is available by primary analyst, stock and coverage universe at <https://www.gs.com/research/hedge.html>.

Not Rated (NR). The investment rating, target price and earnings estimates (where relevant) are removed pursuant to Goldman Sachs policy when Goldman Sachs is acting in an advisory capacity in a merger or in a strategic transaction involving this company, when there are legal, regulatory or policy constraints due to Goldman Sachs' involvement in a transaction, and in certain other circumstances. **Early-Stage Biotech (ES).** An investment rating and a target price are not assigned pursuant to Goldman Sachs policy when this company has neither a drug, treatment or medical device that has passed a Phase II clinical trial nor a license to distribute a post-Phase II drug, treatment or medical device. **Rating Suspended (RS).** Goldman Sachs Research has suspended the investment rating and price target for this stock, because there is not a sufficient fundamental basis for determining an investment rating or target price. The previous investment rating and target price, if any, are no longer in effect for this stock and should not be relied upon. **Coverage Suspended (CS).** Goldman Sachs has suspended coverage of this company. **Not Covered (NC).** Goldman Sachs does not cover this company.

Global product; distributing entities

Goldman Sachs Global Investment Research produces and distributes research products for clients of Goldman Sachs on a global basis. Analysts based in Goldman Sachs offices around the world produce research on industries and companies, and research on macroeconomics, currencies, commodities and portfolio strategy. This research is disseminated in Australia by Goldman Sachs Australia Pty Ltd (ABN 21 006 797 897); in Brazil by Goldman Sachs do Brasil Corretora de Títulos e Valores Mobiliários S.A.; Public Communication Channel Goldman Sachs Brazil: 0800 727 5764 and / or contatogoldmanbrasil@gs.com. Available Weekdays (except holidays), from 9am to 6pm. Canal de Comunicação com o Público Goldman Sachs Brasil: 0800 727 5764 e/ou contatogoldmanbrasil@gs.com. Horário de funcionamento: segunda-feira à sexta-feira (exceto feriados), das 9h às 18h; in Canada by Goldman Sachs & Co. LLC; in Hong Kong by Goldman Sachs (Asia) L.L.C.; in India by Goldman Sachs (India) Securities Private Ltd.; in Japan by Goldman Sachs Japan Co., Ltd.; in the Republic of Korea by Goldman Sachs (Asia) L.L.C., Seoul Branch; in New Zealand by Goldman Sachs New Zealand Limited; in Russia by OOO Goldman Sachs; in Singapore by Goldman Sachs (Singapore) Pte. (Company Number: 198602165W); and in the United States of America by Goldman Sachs & Co. LLC. Goldman Sachs International has approved this research in connection with its distribution in the United Kingdom.

Goldman Sachs International ("GSI"), authorised by the Prudential Regulation Authority ("PRA") and regulated by the Financial Conduct Authority ("FCA") and the PRA, has approved this research in connection with its distribution in the United Kingdom.

European Economic Area: Goldman Sachs Bank Europe SE ("GSBE") is a credit institution incorporated in Germany and, within the Single Supervisory Mechanism, subject to direct prudential supervision by the European Central Bank and in other respects supervised by German Federal Financial Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, BaFin) and Deutsche Bundesbank and disseminates research within the European Economic Area.

General disclosures

This research is for our clients only. Other than disclosures relating to Goldman Sachs, this research is based on current public information that we consider reliable, but we do not represent it is accurate or complete, and it should not be relied on as such. The information, opinions, estimates and forecasts contained herein are as of the date hereof and are subject to change without prior notification. We seek to update our research as appropriate, but various regulations may prevent us from doing so. Other than certain industry reports published on a periodic basis, the large majority of reports are published at irregular intervals as appropriate in the analyst's judgment.

Goldman Sachs conducts a global full-service, integrated investment banking, investment management, and brokerage business. We have investment banking and other business relationships with a substantial percentage of the companies covered by Global Investment Research. Goldman Sachs & Co. LLC, the United States broker dealer, is a member of SIPC (<https://www.sipc.org>).

Our salespeople, traders, and other professionals may provide oral or written market commentary or trading strategies to our clients and principal trading desks that reflect opinions that are contrary to the opinions expressed in this research. Our asset management area, principal trading desks and investing businesses may make investment decisions that are inconsistent with the recommendations or views expressed in this research.

The analysts named in this report may have from time to time discussed with our clients, including Goldman Sachs salespersons and traders, or may discuss in this report, trading strategies that reference catalysts or events that may have a near-term impact on the market price of the equity securities discussed in this report, which impact may be directionally counter to the analyst's published price target expectations for such stocks. Any such trading strategies are distinct from and do not affect the analyst's fundamental equity rating for such stocks, which rating reflects a stock's return potential relative to its coverage universe as described herein.

We and our affiliates, officers, directors, and employees will from time to time have long or short positions in, act as principal in, and buy or sell, the securities or derivatives, if any, referred to in this research, unless otherwise prohibited by regulation or Goldman Sachs policy.

The views attributed to third party presenters at Goldman Sachs arranged conferences, including individuals from other parts of Goldman Sachs, do not necessarily reflect those of Global Investment Research and are not an official view of Goldman Sachs.

Any third party referenced herein, including any salespeople, traders and other professionals or members of their household, may have positions in the products mentioned that are inconsistent with the views expressed by analysts named in this report.

This research is not an offer to sell or the solicitation of an offer to buy any security in any jurisdiction where such an offer or solicitation would be illegal. It does not constitute a personal recommendation or take into account the particular investment objectives, financial situations, or needs of individual clients. Clients should consider whether any advice or recommendation in this research is suitable for their particular circumstances and, if appropriate, seek professional advice, including tax advice. The price and value of investments referred to in this research and the income from them may fluctuate. Past performance is not a guide to future performance, future returns are not guaranteed, and a loss of original capital may occur. Fluctuations in exchange rates could have adverse effects on the value or price of, or income derived from, certain investments.

Certain transactions, including those involving futures, options, and other derivatives, give rise to substantial risk and are not suitable for all investors. Investors should review current options and futures disclosure documents which are available from Goldman Sachs sales representatives or at <https://www.theocc.com/about/publications/character-risks.jsp> and https://www.goldmansachs.com/disclosures/cftc_fcm_disclosures. Transaction costs may be significant in option strategies calling for multiple purchase and sales of options such as spreads. Supporting documentation will be supplied upon request.

Differing Levels of Service provided by Global Investment Research: The level and types of services provided to you by Goldman Sachs Global

Investment Research may vary as compared to that provided to internal and other external clients of GS, depending on various factors including your individual preferences as to the frequency and manner of receiving communication, your risk profile and investment focus and perspective (e.g., marketwide, sector specific, long term, short term), the size and scope of your overall client relationship with GS, and legal and regulatory constraints. As an example, certain clients may request to receive notifications when research on specific securities is published, and certain clients may request that specific data underlying analysts' fundamental analysis available on our internal client websites be delivered to them electronically through data feeds or otherwise. No change to an analyst's fundamental research views (e.g., ratings, price targets, or material changes to earnings estimates for equity securities), will be communicated to any client prior to inclusion of such information in a research report broadly disseminated through electronic publication to our internal client websites or through other means, as necessary, to all clients who are entitled to receive such reports.

All research reports are disseminated and available to all clients simultaneously through electronic publication to our internal client websites. Not all research content is redistributed to our clients or available to third-party aggregators, nor is Goldman Sachs responsible for the redistribution of our research by third party aggregators. For research, models or other data related to one or more securities, markets or asset classes (including related services) that may be available to you, please contact your GS representative or go to <https://research.gs.com>.

Disclosure information is also available at <https://www.gs.com/research/hedge.html> or from Research Compliance, 200 West Street, New York, NY 10282.

© 2026 Goldman Sachs.

You are permitted to store, display, analyze, modify, reformat, and print the information made available to you via this service only for your own use. You may not resell or reverse engineer this information to calculate or develop any index for disclosure and/or marketing or create any other derivative works or commercial product(s), data or offering(s) without the express written consent of Goldman Sachs. You are not permitted to publish, transmit, or otherwise reproduce this information, in whole or in part, in any format to any third party without the express written consent of Goldman Sachs. This foregoing restriction includes, without limitation, using, extracting, downloading or retrieving this information, in whole or in part, to train or finetune a machine learning or artificial intelligence system, or to provide or reproduce this information, in whole or in part, as a prompt or input to any such system.